

Contact

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www.linkedin.com/in/muhammad-faizan-ahmed (LinkedIn)
engr-muhammad-faizan-ahmed.github.io/portfolio/ (Portfolio)

Top Skills

RoboDK
Robot Simulation
Node-RED

Languages

English (Professional Working)
Urdu (Native or Bilingual)

Certifications

Introduction to Mechanical Engineering Design and Manufacturing with Fusion 360
Take your resume to the next level with Canva
Generative AI: Prompt Engineering Basics
AI For Everyone
Programming for Everybody (Getting Started with Python)

Honors-Awards

Silver Medalist
Gold Medalist
Allan and Nesta Ferguson Masters Scholarship

Engr. Muhammad Faizan Ahmed

Postgraduate Robotics and Automation | Automation Engineer @ Industronics Engineering | Robotics, Industrial Automation & AI Enthusiast - exploring emerging tech
United Kingdom

Summary

I am an MSc Robotics & Automation student at Manchester Metropolitan University with hands-on experience in industrial automation, machine vision, and robotic systems integration. My work focuses on designing and deploying practical solutions for real-world production environments, combining control systems, software, and intelligent automation. I have professional experience as an Automation Engineer, where I developed a full SCADA system using WinCC Unified for an automated warehouse and implemented a vision-based cloth width measurement system using computer vision techniques. This experience strengthened my ability to work across the full automation stack; from PLC programming and HMI development to machine vision and system integration. My technical skill set includes PLC programming (Siemens, Delta), SCADA development, industrial robotics, machine vision (OpenCV, YOLO), and simulation tools such as RoboDK and CoppeliaSim. I also have a strong foundation in Python, C++, embedded systems, and hardware design, including PCB development and 3D modelling. I am particularly interested in robotics and smart manufacturing, with a focus on autonomous systems, industrial inspection, and collaborative robotics. My current MSc project involves cobot synchronisation for industrial sewing applications, where I am working on coordinated robotic control for improved efficiency and precision. I am driven to contribute to the future of Industry 4.0 by developing efficient, scalable, and intelligent automation solutions. I am actively seeking opportunities in robotics and automation engineering where I can apply my skills in system design, integration, and deployment while continuing to grow in a challenging industrial environment.

Experience

Industronics Engineering

6 months

Automation Engineer

June 2025 - August 2025 (3 months)

Karāchi

Junior Automation Engineer

March 2025 - May 2025 (3 months)

Karāchi, Sindh, Pakistan

Upwork

Freelancer

July 2022 - August 2025 (3 years 2 months)

I am providing both technical and non-technical services. Technical services include Digital Logic Designing, Python, MATLAB, and Fusion 360. While non-technical services include Logo Designing, data entry etc.

National University of Sciences and Technology (NUST)

Research Officer - Hardware Design Engineer

March 2024 - March 2025 (1 year 1 month)

Karāchi, Sindh, Pakistan

- Worked on the design, prototyping, and testing of drones, including quadcopters and high-speed targeted drones.
- Worked on PLC-based systems such as AV bird launcher, PLC Test Bench etc., and also contributed to HMI designing.
- Designed PCBs from schematic to 3D models using Altium and EasyEDA, with 3D modeling and custom enclosure design using SolidWorks.
- Contributed to a radar simulator project by developing a user-friendly frontend interface using PyQt6 and integrating software components in Linux environments.

Industry Innovative Engineers - IIEngineers

PLC Intern

November 2024 - December 2024 (2 months)

Learned the basics of PLCs, Communication Protocols, Topologies, and basic ladder logic programming on the TIA Portal used in industrial automation

Indus Technetronic

IoT Engineer

July 2023 - December 2023 (6 months)

Hyderabad, Sindh, Pakistan

- Prototyping IoT devices.
- Designing PCBs.
- Developing 3d-Printed Enclosures for IoT devices.

Pakistan Aeronautical Complex Kamra

Mechatronics Engineering Intern

June 2023 - June 2023 (1 month)

Punjab, Pakistan

- Learned about Manufacturing processes of JF-17 Thunder & MUSHAQ Jet.
- Explored different manufacturing workshops.

AMROY

Junior CAM Engineer (Internee)

July 2022 - August 2022 (2 months)

Karāchi, Sindh, Pakistan

- Producing thorough documentation of Electronics components with their specifications in line with company requirements.
- Understanding the different processes involved in manufacturing PCBs.
- In-depth understanding of PCB layers and internal architecture.

ZIMCO International Corporation

IOT and Power System Intern

January 2022 - February 2022 (2 months)

Pakistan

In this internship Program, I learned regarding Power Transmission, Distribuiton. I also learned regarding designing our own Main Electric Panel for domestic applications.

Automation Play

Industry 4.0 and Industrial Automation Internee

October 2021 - November 2021 (2 months)

NCAI - NEDUET, Smart City Lab

Computer Vision Internee

December 2020 - January 2021 (2 months)

Karāchi, Sindh, Pakistan

It was a nice experience where I was to perform image dataset labeling(Annotation) for Weed Robot. I am thankful to Sir Imtiaz who guided me throughout.

Education

The Manchester Metropolitan University

Master of Science - MS, Robotics and Automation · (September 2025 - September 2026)

Mehran University of Engineering and Technology

Bachelor's degree, Mechatronics, Robotics, and Automation Engineering · (October 2019 - December 2023)